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EXECUTIVE SUMMARY:

The Sentinel Framework is a vision for quantum engineers dealing with the cusp of algorithms and hardware. From cold atom gravimeters to unclonable hardware fingerprints. So, our critical infrastructure is protected by gravity locked security nodes and hardware root of trust – built by our own engineers, on our own terms.

The world stands at the threshold of the quantum era. The Q-Day is not a distant phenomenon; it is here. Nations such as the US, China, and the EU are investing billions to stay ahead in the quantum leap. Quantum hardware, cryptography, and sensing are moving decisively beyond theoretical simulation toward physical sovereignty.

Pakistan is not exceptional to this transition. We can either import quantum talent or import hardware, or we can develop our own workforce capable of designing, building, and securing quantum systems.

In response to the National Center for Quantum Computing (NCQC) call for curriculum ideas, this proposal **The Sentinel Framework** is a specialized track for the BS in Quantum Computing (BSQC) and the MS in Quantum Systems Engineering (MS-QSE).

The very idea for the Sentinel Framework comes from validated work: the RAQT protocol tested on IBM and Chinese quantum hardware and the “Quantum for All” 4-tier curriculum (IBM QAMP 2025).

The focus on software and hardware interfacing is the new approach adopted here, infrastructure resilience and robust capabilities, considering that the physical layer is where sovereignty begins. This is where our engineers can leap ahead.

The Sentinel Mandate is a three-part roadmap:

- (1) integrate PUF, metrology, and secure network electives into the BSQC program;
- (2) establish two specialized labs (MOT/UHV for sensing, AWG/cryo for PUF);
- (3) train faculty within 18 months.

This roadmap aligns with NCQC’s goal of producing industry-ready quantum engineers.

2. Existing Validation: Proven Foundation

The Sentinel Framework is built on real work that I have already done.

RAQT Protocol – Tested on IBM and a Chinese Quantum Computer and NetSquid Simulation

RAQT stands for Robust Anonymous Quantum Transmission. It is my implementation of the Christandl-Wehner 2004 protocol. The goal is simple: send one secret bit across a quantum network without revealing who sent it. You use GHZ states, Z-gate encoding, and X-basis measurements. Then you XOR the results. That gives you the bit.

I coded RAQT from scratch. First in Qiskit, then in NetSquid. Then I took it to real hardware.

- **IBM Quantum** – the protocol worked. Ideal simulation gave 100% success. With 1% depolarising noise, it still succeeded 97% of the time.
- **Origin Wuyuan** – that is a quantum computer in China. I ran the same protocol there. It worked too. Cross-platform validation.

The code, the simulation scripts, and the noise analysis are all on my GitHub. This work directly feeds into the **Secure Quantum Networks** elective I am proposing. It also proves that our students can go beyond simulators and run code on real machines, even on hardware from different countries.

“Quantum for All” Curriculum – IBM QAMP 2025

“Quantum for All” is a 4-tier quantum curriculum. I proposed it and led the project as part of IBM QAMP (Qiskit Advocate Mentorship Program) 2025.

Here are the tiers:

- **Tier 1 (grades 6-8):** stories, games, intuition. No hard math. Just curiosity.
 - **Tier 2 (grades 9-11):** Bloch sphere, basic probability, visualisations.
 - **Tier 3 (undergraduate):** Qiskit programming, formal notation, simple algorithms.
 - **Tier 4 (expert):** advanced protocols – no-cloning, EPR pairs, QKD, error mitigation.
-

The curriculum is aligned with STEM subjects. Physics teachers can teach quantum through physics. Math teachers through matrices. Computer science teachers through logic gates. I also added multilingual support – Roman English and Spanish. The pedagogy uses Bloom’s Taxonomy and the VAK learning model (visual, auditory, kinesthetic).

We delivered everything on time. All pull requests were completed. The full framework is documented.

Why does this matter for the Sentinel Framework? Because it gives us a ready-to-use pedagogical model for the BSQC program.

3. BS in Quantum Computing (BSQC): Core Curriculum

Program Narrative & Career Pathways

The BS in Quantum Computing at NCQC is a four-year degree (130 credit hours) that trains engineers to understand quantum systems from the ground up – from the mathematical foundations of Hilbert spaces to the physical implementation of qubits on real hardware. The programme integrates AI/ML throughout, recognising that the coming decade will demand graduates who can both use AI to control quantum hardware and develop quantum algorithms to accelerate AI.

Global benchmarks inform this design.

Tsinghua University’s “Yao Class” founded by Turing Award winner Andrew Chi-Chih Yao has, since 2005, nurtured undergraduates to compete with students from MIT, Stanford and Princeton.

It offers three specialised tracks (Computer Science, Artificial Intelligence, and Quantum Information) with a tiered model: the first two years focus on intensive foundational training, the final two years on deep specialisation.

A comparable structure is found at the University of Science and Technology of China (USTC), whose quantum information science programme combines physics fundamentals (Quantum Mechanics I, Solid State Physics) with specialised professional courses in quantum communication, quantum computation and quantum precision measurement.

In the United States, Harvard’s Quantum Science & Engineering PhD programme sits at the interface of physics, chemistry and engineering, providing students with opportunities to explore the realisation and application of fully engineered quantum systems.

MIT offers a sequence of graduate-level quantum information science courses that cover density-matrix formalism, operator-sum representations, fault-tolerance and quantum complexity theory. UC Berkeley now even offers a course in which students design and test quantum chips – a first-of-its-kind hands-on hardware experience.

Proposed Scheme of Study (130 Credit Hours)

The curriculum follows the generic structure for computing degree programmes approved by the Higher Education Commission of Pakistan and the Washington Accord:

Area	Credit Hours	Courses
Major (Computing) Core	46	14
Quantum Core	18	6
Domain Electives (including Sentinel Track)	21	7
Mathematics & Supporting Courses	12	4
Elective Supporting (Professional Certification)	3	1
General Education Requirement	30	12
Total	130	44

This distribution mirrors international standards. USTC’s quantum information science BS also requires approximately 130–140 credits, combining physics foundations with AI and machine learning electives. Tsinghua’s Yao Class follows a “strong foundational knowledge + interdisciplinary exploration” model, with specialisations chosen after the first academic year.

Sentinel-Integrated Semester Plan

The three Sentinel electives: **Hardware Root of Trust (PUF)**, **Atomic Metrology & Gravimetry**, **Secure Quantum Network** are distributed across Semesters 5–7. Each elective now includes explicit AI/ML modules that reflect the latest global trends (e.g., neural-network quantum error correction, reinforcement learning for quantum control, and AI-driven anomaly detection for quantum networks).

Year	Semester	Core Courses	Sentinel Elective (Choose One)	AI/ML Integration
1	Fall	• Programming Fundamentals for QC, • QR1 (Discrete Structures), • ICT • Functional English	–	–
1	Spring	• Object Oriented & Scientific Programming, • Database Systems, • Digital Logic Design, • Multivariable Calculus, Linear Algebra	–	–
2	Fall	• Data Structures & Complexity, • Information Security, • AI, • Computer Networks, Software Engineering, Probability & Statistics	–	AI Fundamentals (core AI course)
2	Spring	• Computer Organisation & Assembly Language, • Domain Core (Linear Algebra for QC), • Domain Core (Quantum Programming – Qiskit/Cirq),	–	–

Year	Semester	Core Courses	Sentinel Elective (Choose One)	AI/ML Integration
		- Natural Sciences (Applied Physics), - Expository Writing, - Islamic Studies		
3	Fall	- Operating Systems, Domain Core: (Quantum Information Theory), Domain Core: (Quantum Mechanics for Computing), - GE: Social Science	Hardware Root of Trust (PUF)	AI for Hardware Security – ML-based PUF modelling and attack simulation
3	Spring	Domain Core: (Quantum Algorithms), Domain Core: (Parallel & Distributed Quantum Computing), - Domain Elective:(×2), - GE Elective	Atomic Metrology & Gravimetry	AI for Quantum Control – reinforcement learning for cold-atom systems
4	Fall	- Final Year Project – I, Analysis of Algorithms (Classical & Quantum), - Domain Elective, - Technical & Business Writing, - Entrepreneurship	Secure Quantum Networks	AI for Network Security – anomaly detection in QKD
4	Spring	- Final Year Project – II, - Ideology & Constitution of Pakistan, - Arts & Humanities (Professional Practices), - Civics & Community Engagement	–	Capstone AI-Quantum Project

AI/ML Integration in Sentinel Electives

Each Sentinel elective now includes a dedicated AI/ML module, reflecting global rdirections:

Elective	AI/ML Module	Global Precedent
Hardware Root of Trust (PUF)	• ML-based PUF modelling; training classifiers to distinguish real from modelled PUFs; • AI-driven attack simulation	• MIT “Secure Hardware Design” uses ML for side-channel analysis; • Harvard QSE programme explores AI-assisted hardware characterisation
Atomic Metrology & Gravimetry	• Reinforcement learning for cold-atom control; • AI-based optimisation of MOT loading; • neural-network-assisted real-time noise cancellation	• ETH Zurich teaches RL for quantum control; • Stanford APPPHYS 228 (Quantum Hardware) includes AI for quantum sensing; • TU Delft has a dedicated course on RL for quantum devices
Secure Quantum Networks	• AI-based anomaly detection for QKD networks; • deep learning for eavesdropper localisation; • neural-network error correction for quantum repeaters	• QuTech (Delft) runs AI-assisted QKD post-processing; • Harvard QSE includes quantum network security modules; • MIT 6.4530 covers ML for quantum error mitigation

The AI-Driven Advantage

The AI/ML components are fully integrated into the Sentinel electives. This mirrors the approach of leading programmes:

Tsinghua’s Center for Quantum Information conducts research on “Quantum artificial intelligence, quantum information and computation, topological phases of matter” and physical implementations (ion traps, neutral atoms, superconducting circuits).

Harvard’s Quantum Science and Engineering initiative explicitly trains researchers who need “deep knowledge of science **and** engineering principles”, a dual competence that the Sentinel AI modules help to build.

Berkeley’s quantum chip design course proves that hardware-level education is now feasible at the undergraduate level.

The proposed plan an **enhancement** of the existing NCQC BSQC scheme, adding three specialised electives and their associated labs, just as Tsinghua adds specialisations after the

first academic year, and just as top US universities offer quantum tracks within broader engineering degrees.

4. MS in Quantum Systems Engineering (MS-QSE):

Advanced Infrastructure

The MS in Quantum Systems Engineering (MS-QSE) is a two-year (30-credit) programme designed for students who already have a BS in physics, computer science, electrical engineering, or a related field. The goal is to move graduates from pure simulation to hands-on hardware interfacing, infrastructure design, and security engineering. The curriculum is built around two tiers: a computational foundation that leverages HPC and the NCQC laboratory, and two strategic research nodes, Atomic Metrology & Gravimetry, and Hardware Characterisation & Root of Trust.

The Transition from Simulation to Hardware Interfacing:

A real quantum engineer must be able to connect a cryostat, tune a laser, or extract a PUF response from a raw chip. The MS-QSE closes that gap.

The programme recognises that Pakistan cannot import every quantum technician. We must train our own people to maintain, calibrate, and secure quantum hardware. The transition is organised in three phases:

Phase	Focus	Location	Example Activity
1	Computational foundations	HPC cluster, NCQC lab	• Simulate a 10-qubit GHZ state on a high-performance node; • benchmark Qiskit vs Cirq
2	Hardware interfacing	Atomic Metrology Lab (MOT/UHV)	• Build a simulated magneto-optical trap; • analyse real gravimeter data
3	Root-of-trust engineering	PUF Lab (AWG/cryo)	• Characterise simulated PUF responses; • design an authentication protocol for a quantum network

Each phase includes a lab component that uses real equipment where available, and high-fidelity simulation where hardware is not yet deployed. This mirrors the approach of leading programmes such as the **MIT Quantum Engineering** track, the **QuTech MSc** (Delft), and the **ETH Zurich Quantum Engineering** master.

Tier 1: Computational Foundations – HPC & NCQC Laboratory

The first tier provides students with the computational infrastructure needed to design and test quantum protocols at scale. It is built around two facilities:

A. High-Performance Computing (HPC) Cluster

- **Purpose:** Run large-scale simulations of quantum circuits, noise models, and quantum networks (up to 30–40 qubits using state-vector simulators; many more using tensor-network methods).
- **Infrastructure:** Existing HPC nodes at UET Lahore/Narowal, augmented with GPU accelerators (NVIDIA A100 or equivalent) for machine-learning-assisted quantum simulations.
- **Software stack:** Qiskit, Cirq, NetSquid, PennyLane, and custom Python libraries for AI-driven error mitigation.

B. NCQC Quantum Computing Laboratory

- **Purpose:** Provide cloud access to real quantum hardware (IBM Quantum, Origin Wuyuan, and future local testbeds) and a local emulation environment.
- **Infrastructure:** Dedicated workstations with Jupyter hubs, version control (Git), and a shared repository of verified quantum circuits.
- **Learning outcomes:** Students learn to submit jobs to remote quantum processors, analyse measurement results, and account for noise in real hardware.

Courses in Tier 1 (two core courses):

Course	Description	Lab Component
Advanced Quantum Simulation	State-vector, density matrix, and tensor-network methods; noise modelling; parallelisation on HPC clusters.	Run a 25-qubit random circuit simulation on HPC; benchmark against a real 5-qubit IBM device.

Course	Description	Lab Component
Quantum Computing with Real Hardware	Transpilation, calibration, error mitigation; accessing IBM and Origin hardware via APIs; handling limited qubit connectivity.	Implement the RAQT protocol on IBM Quantum and on Origin Wuyuan; compare success rates.

These courses are directly informed by the **USTC quantum information science** programme and the **Harvard QSE** core.

Tier 2: Strategic Research Nodes

The second tier consists of two specialised research nodes, each equipped with simulation tools and (in future) physical hardware. Students choose one node as their primary specialisation.

A. Atomic Metrology & Gravimetry Lab (MOT/UHV)

Mission: Train engineers to build and operate cold-atom systems for precision sensing – gravimeters, gyroscopes, and magnetometers. These are the building blocks of gravity-locked navigation (GPS-denied environments).

Infrastructure (planned):

- Magneto-Optical Trap (MOT) kits (e.g., from ColdQuanta, TOPTICA, or home-built).
- Ultra-High Vacuum (UHV) chamber with ion pump and titanium sublimation pump.
- Tunable diode lasers (780 nm for rubidium, or 589 nm for sodium).
- CCD camera and photodetectors for fluorescence imaging.
- Control electronics (PID controllers, arbitrary waveform generators).

Learning outcomes:

- Design and align a MOT (simulated initially, then on a table-top setup).
- Measure the temperature of trapped atoms via time-of-flight imaging.

- Operate a Mach–Zehnder atom interferometer and extract gravity acceleration g .
- Analyse noise sources (vibration, magnetic fields, laser frequency drift).

Laboratory exercises (simulated + hardware):

Lab	Activity	Tools
1	Simulate laser cooling: solve optical Bloch equations for a two-level atom.	Python (QuTiP)
2	Design a MOT magnetic field profile (anti-Helmholtz coils) using Biot-Savart.	Python / COMSOL (basic)
3	Process real gravimeter data from an open dataset (e.g., iXblue) to extract g .	Python (pandas, scipy)
4	(Advanced) Build a small-scale rubidium MOT demonstration (if funding allows).	Physical hardware

Connection to Sentinel Framework:

This node directly supports the **Atomic Metrology & Gravimetry** elective in the BSQC programme and provides a research environment for MS theses.

Global equivalents:

MIT’s “Atomic Physics” lab course, ETH Zurich’s “Quantum Sensors” practical, and the **QUESTING** doctoral network (European quantum internet training) all include cold-atom gravimetry modules.

5. Technical Accreditation & Global Benchmarking

A modern quantum engineering programme must be recognised by professional bodies and aligned with international standards. The proposed BSQC and MS-QSE programmes are designed to meet the accreditation requirements of the **Pakistan Engineering Council (PEC)** and to be consistent with the **Washington Accord** for mutual recognition of engineering degrees.

Accreditation Strategy

Level	Target	Rationale
BSQC	PEC (Pakistan Engineering Council)	All engineering programmes in Pakistan must be accredited by PEC for graduates to register as professional engineers. The curriculum follows the HEC-approved generic structure for computing degrees.
MS-QSE	Not separately accredited	Master's programmes typically do not require engineering accreditation; however, the degree will be recognised by HEC and international universities for PhD admission.

Alignment with Washington Accord Graduate Attributes

The Washington Accord defines 12 graduate attributes for professional engineers. The Sentinel-enhanced BSQC programme explicitly addresses each attribute. Key examples:

Attribute	How BSQC Delivers	Sentinel Contribution
Engineering Knowledge	• Mathematics, • physics, • quantum mechanics, • computing core	• PUF, • metrology, • secure networks provide hardware-level knowledge
Problem Analysis	• Quantum algorithms, • error correction,	AI/ML modules for anomaly detection in QKD

Attribute	How BSQC Delivers	Sentinel Contribution
	noise modelling	
Design/Development of Solutions	Capstone projects, lab courses	Sentinel electives require designing a PUF-authenticated network or a cold-atom gravimeter simulation
Investigation	Final-year research project	Thesis-level investigation in MS-QSE
Modern Tool Usage	- Qiskit, - Cirq, - NetSquid, - HPC clusters	Hands-on with real IBM and Origin hardware
Ethics & Security	- Information Security course, - Professional Practices	Hardware root of trust and quantum network security modules explicitly address security ethics

Benchmarking Against International Programmes

To ensure global competitiveness, the curriculum has been benchmarked against the following programmes (as detailed in previous sections):

Institution	Programme	Key Features Adopted
Tsinghua University	Yao Class (Quantum Information)	Tiered specialisation after foundational year; strong AI integration
USTC	BS in Quantum Information Science	Broad physics foundation; professional electives in sensing, materials, and devices

Institution	Programme	Key Features Adopted
MIT	Quantum Engineering (graduate track)	HPC simulation + hardware interfacing labs
QuTech (Delft)	MSc Quantum Information Science & Technology	NetSquid simulation; quantum internet focus
Harvard	Quantum Science & Engineering	Transition from physics to engineering; hardware-focused courses
Stanford	APPPHYS 228 (Quantum Hardware)	Quantum sensing, control, and hardware characterisation
ETH Zurich	Quantum Engineering MSc	Cold-atom sensing; quantum control with AI

The Sentinel Framework’s three electives directly correspond to courses taught in these top programmes, ensuring that NCQC graduates are competitive for international research positions and industry roles.

Professional Certifications

Students are encouraged to earn industry-recognised certifications as part of their degree:

Certification	Provider	Relevance
Qiskit Global Summer School Certificate	IBM Quantum	Foundational and advanced quantum programming
Cirq / TensorFlow Quantum Training	Google	Quantum machine learning

Certification	Provider	Relevance
NetSquid Tutorial Certificate	QuTech	Quantum network simulation
Quantum Hardware Security (future)	NCQC / Sentinel	Custom certification for PUF and metrology

These certifications are listed as “Elective Supporting Courses” (3 credits) in the BSQC scheme of study.

6. Faculty Development & Training Roadmap

The success of the Sentinel Framework depends on having faculty who are confident teaching quantum hardware, security, and AI/ML integration. Currently, Pakistan has a very small pool of quantum-trained academics. This roadmap outlines a phased approach to build that capacity.

Phase 1: Foundations & Tool Fluency (Months 1–6)

Goal: Enable faculty to teach the core quantum courses (algorithms, programming, cryptography) using existing simulators and cloud-based hardware.

Activity	Deliverable	Partners / Resources
Train the trainer workshops (online)	Faculty complete Qiskit Global Summer School and Cirq tutorials	IBM Quantum, Google Quantum AI
NetSquid training for quantum networks	Faculty can simulate RAQT and basic QKD networks	QuTech (online resources)
Establish peer-learning group	Weekly journal club and code-review session	NCQC internal

Expected outcome: 5–8 faculty members proficient in teaching BSQC core courses.

Phase 2: Specialised Hardware Operations (Months 7–12)

Goal: Prepare faculty to teach the Sentinel electives (PUF, Atomic Metrology, Secure Quantum Networks) and to supervise student labs.

Elective	Training Activity	Resource Provider
PUF (Hardware Root of Trust)	Online course “Secure Hardware Design” (MIT OpenCourseWare) + lab tutorials from QuantumPUF project	MIT OCW, EIC Pathfinder
Atomic Metrology	Virtual lab on cold-atom systems (e.g., QuVis simulations) + guest lecture from a cold-atom researcher	University of Strathclyde, ETH Zurich (recorded lectures)

Elective	Training Activity	Resource Provider
Secure Quantum Networks	NetSquid advanced tutorials; hands-on with IBM Quantum's Qiskit Runtime	QuTech, IBM Quantum

Expected outcome: 3 - 4 faculty members can each teach one Sentinel elective and supervise student projects.

Phase 3: Deep-Tech Innovation & Global Research (Months 13–18)

Goal: Enable faculty to conduct original research in quantum infrastructure, publish in international journals, and attract research funding.

Activity	Deliverable	Partners
Faculty exchange / virtual visiting scholar	Joint research project with QuTech, MIT, or Tsinghua	NCQC, HEC (funding)
Establish a research group in quantum security	First student publications on PUF or RAQT variants	Sentinel Framework
Apply for international grants (EU, ICTP, OQI)	At least one funded project proposal submitted	CERN OQI, QIA

Expected outcome: NCQC becomes a recognised research node in quantum infrastructure security.

Timeline Summary

Phase	Duration	Key Milestone
Phase 1	Months 1–6	All core faculty certified in Qiskit/Cirq

Phase	Duration	Key Milestone
Phase 2	Months 7–12	Three Sentinel electives taught for the first time
Phase 3	Months 13–18	First research publication and grant application submitted

Budget Estimate (Faculty Development)

Item	Estimated Cost (PKR)	Source
Online course subscriptions (Coursera, edX)	Subject to Currency Rate	HEC / NCQC
Visiting faculty honorarium (virtual)	Subject to Currency Rate	NCQC
Conference travel (1 international / year)	Subject to Currency Rate	HEC / PSF
Laboratory equipment (MOT demo kit, AWG, cryo-amplifier)	Subject to Currency Rate	PSDP / special grant

Total approximate: Modest compared to the cost of importing hardware expertise.

7. Implementation Timeline & Estimated Investment

Milestone	Duration	Rough Cost (PKR)
Procurement of MOT kit, UHV chamber, lasers	Subject to Situation	Subject to Currency Rates
Procurement of AWG, cryogenic amplifiers	Subject to Situation	Subject to Currency Rates
Faculty training (workshops, online courses)	Subject to Situation	Subject to Currency Rates
Full Sentinel Track launch	Subject to Situation	Subject to Currency Rates

Risk mitigation: Import delays, use simulated labs initially; lack of trained faculty, partner with international online courses and virtual guest lectures.

8. Concluding Statement: Physics of Trust

The Sentinel Framework is a concrete, evidence-based roadmap to equip Pakistan's quantum workforce with the skills to secure national infrastructure, authenticate quantum devices, and sense the environment with cold-atom precision. The framework builds on already-validated work – RAQT, “Quantum for All”, and the PUF module – and aligns with the best practices of Tsinghua, MIT, QuTech, and Harvard.

By adopting the Sentinel electives, establishing the two specialised labs, and investing in faculty development, NCQC will produce engineers who can protect Pakistan's digital sovereignty at the physical layer. This is where the quantum race will be won or lost. And this is where Pakistan can leap ahead not by importing technology, but by building our own capability, on our own terms.

Respectfully submitted,

Noor Ul Ain Faisal

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Lead, “Quantum for All” Curriculum (IBM QAMP 2025)

GitHub: [learningdungeon](#)

Repository: [qamp-2025](#)